PHASE 3 • PROJECT DESIGN PHASE

✓ VERIFIED SPECIFICATION

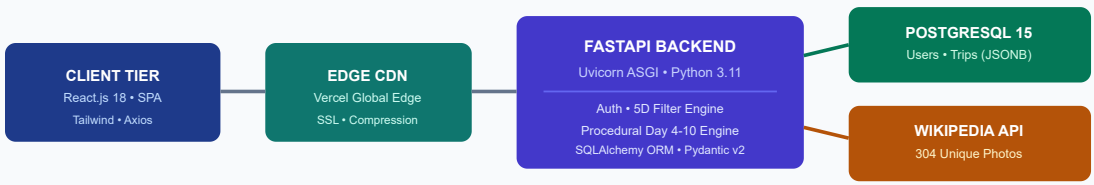
Solution Architecture

AI-Powered Travel Planner • Smart Situational Itinerary Engine

Live URL: travel-planner-iota-plum.vercel.app **Repo:** github.com/itskian2007/travel-planner **Tech:** React + FastAPI + PostgreSQL **Team:** itskian2007

1. Full End-to-End System Architecture

The system employs a production-grade 3-tier client-server topology featuring a single-page application (SPA) frontend served via global edge content delivery networks, communicating over authenticated RESTful APIs to an asynchronous ASGI backend backed by a relational database and external media feeds.



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2. Frontend Component Hierarchy Tree

REACT COMPONENT HIERARCHY — FRONTEND/SRC/

```
App.jsx (Root Application Component)
 ├── AuthContext.jsx (Global Authentication & JWT State Provider)
 └── BrowserRouter (React Router v6 SPA Routing)
      ├── Navbar.jsx (Responsive Navigation Header & Auth Status)
      ├── Routes:
      │    ├── / (LandingPage.jsx - Hero Banner, Features, Value Proposition)
      │    ├── /login (LoginPage.jsx - JWT Email/Password Authentication)
      │    ├── /register (RegisterPage.jsx - User Signup Workflow)
      │    ├── /forgot-password (ForgotPasswordPage.jsx - Self-Service Recovery)
      │    ├── /home (HomePage.jsx - 20 Destination Grid)
      │    │    ├── FilterPanel.jsx (Mood, Budget, Diet, Persona Selectors)
      │    │    └── DestinationCard.jsx × 20 (Photo, Badges, Meta, Click Handler)
      │    └── /itinerary/:id (ItineraryPage.jsx - Full 10-Day Plan)
      │         ├── DayCard.jsx × 10 (Expandable Accordion Cards)
      │         │    ├── AttractionCard.jsx (Title, Description, Duration)
      │         │    ├── LandmarkImage.jsx (Wikipedia Verified Image Fetcher)
      │         │    └── DiningCard.jsx (Filtered Lunch & Dinner Recommendations)
      └── Footer.jsx (Copyright, Legal, Documentation Links)
```

3. Backend Modular Package Organization

MODULE / FILE PATH	PRIMARY ARCHITECTURAL RESPONSIBILITY	KEY DEPENDENCIES & EXPORTS
`backend/main.py`	FastAPI App Initialization, CORS, Router Mounting	FastAPI, Uvicorn, Starlette Middleware
`backend/routers/auth.py`	Registration, Login, JWT Token Issuance	bcrypt, python-jose, OAuth2PasswordBearer
`backend/routers/destinations.py`	Catalog Search, 5D Filtering, City Details	destinations_db.py, Pydantic Schemas
`backend/routers/itinerary.py`	Hybrid Day 1-3 Curated + Day 4-10 Procedural	Procedural Engine, Wikipedia Image Service
`backend/models.py`	SQLAlchemy Relational Database Schema Definitions	DeclarativeBase, ForeignKey, Column Types